

Parameter Summary

TABLE I
SYSTEM, FLEXIBLE-RESOURCE, AND ROBUST-OPTIMIZATION PARAMETERS

Parameter	Symbol	Value
Scheduling horizon	T	24 h
Scheduling interval	Δt	1 h
Number of AGVs	N_{AGV}	20
Number of reefer containers	N_{RC}	100
Number of berthed vessels	N_{VES}	3
Available charging piles	N_{piles}	10
AGV battery capacity	E_{cap}	100 kWh
AGV transportation energy per period	$E_{consume}$	5 kWh
TOU electricity price (off-peak/flat/peak)	$\pi_{elec,t}$	0.456/0.816/1.44 yuan/kWh
Fuel price	π_{fuel}	0.76 yuan/kWh
MR carbon-dioxide consumption coefficient	χ^{MR}	0.75
CCS carbon-capture efficiency	χ_{CCS}	0.40
Electric-load shedding penalty	λ_e	25 yuan/kWh
Heat-load shedding penalty	λ_h	15 yuan/kWh
Cooling-load shedding penalty	λ_c	15 yuan/kWh
Uncertain-variable set	$\mathcal{N}$	{PV, WT, VES}
Maximum forecast-deviation ratio	$\Theta_{n,t} / \bar{P}_{n,t}^e$	20%
Temporal uncertainty budgets	$\Psi_{PV}, \Psi_{WT}, \Psi_{VES}$	6, 6, 6
Critical fault interval	$\mathcal{T}_f^{out}$	9–14 h

TABLE II
PORT-SIDE EQUIPMENT AND ENERGY-STORAGE PARAMETERS

Device	Output limit $P_{s,max}$ (kW)	Conversion efficiency	Ramp limit $P_{s,max}^r$ (kW)
GT	4500	$(\eta_e, \eta_h) = (0.30, 0.60)$	400
HP	1000	$\eta_{HP} = 0.96$	200
ER	1000	$\eta_{ER} = 3.00$	250
HFC	450	$(\eta_e^{HFC}, \eta_h^{HFC}) = (0.55, 0.40)$	60
EL	300	$\eta^{EL} = 0.88$	60
Grid	4500	–	–

Storage	Energy range $[C_{min,m}^{ESS}, C_{max,m}^{ESS}]$ (kWh)	Power limit $P_{m,max}$ (kW)	Efficiency $\eta_{m,ch}/\eta_{m,dis}$	Initial energy $C_{1,m}^{ESS}$ (kWh)
EES	[300, 1500]	300	0.85/0.95	750
TES	[200, 1200]	280	0.98/0.98	600
HES	[300, 1000]	250	0.90/0.85	500
GES	[200, 1200]	230	0.95/0.95	600

TABLE III
VESSEL-SIDE PARAMETERS

Parameter	Symbol	V1	V2	V3
Arrival time (h)	τ_v^{arr}	1	1	1
Departure time (h)	τ_v^{dep}	24	24	24
Maximum charging power (kW)	$P_{v,max}^{ves,ch}$	288	296	306
Maximum discharging power (kW)	$P_{v,max}^{ves,dis}$	290	298	312
Charging efficiency	$\eta_{v,ch}^{ves}$	0.944	0.948	0.952
Discharging efficiency	$\eta_{v,dis}^{ves}$	0.945	0.949	0.953
Minimum stored energy (kWh)	$C_{v,min}^{ves}$	190	195	205
Maximum stored energy (kWh)	$C_{v,max}^{ves}$	2880	2960	3060
Arrival stored energy (kWh)	$C_v^{ves,arr}$	1440	1480	1530

TABLE IV
C&CG, UNCERTAINTY, COST, AND CARBON PARAMETERS

Parameter	Manuscript symbol	Code value
Maximum C&CG iterations	$m_{\max}$	20
Absolute convergence tolerance	ε	1.0×10^{-5}
PV uncertainty budget	Ψ_{PV}	6
WT uncertainty budget	Ψ_{WT}	6
VES-load uncertainty budget	Ψ_{VES}	6
PV maximum deviation ratio	$\Theta_{\text{PV},t} / \bar{P}_{\text{PV},t}^e$	0.20
WT maximum deviation ratio	$\Theta_{\text{WT},t} / \bar{P}_{\text{WT},t}^e$	0.20
VES-load maximum deviation ratio	$\Theta_{\text{VES},t} / \bar{P}_{\text{VES},t}^e$	0.20
Grid-electricity emission factor	μ_e	0.0007
Carbon conversion coefficient	ω_c	1.0
Default shedding penalties (electric/heat/cooling)	$(\lambda_e, \lambda_h, \lambda_c)$	(25, 15, 15) yuan/kWh

TABLE V
AGV AND REEFER-CONTAINER PARAMETERS

Parameter	Manuscript symbol	Code value
AGV number	N_{AGV}	20
Maximum AGV charging power	$p_{\max}^{\text{ch}}$	50 kW
Maximum AGV discharging power	$p_{\max}^{\text{dis}}$	50 kW
AGV charging/discharging efficiency	$\eta^{\text{ch}} / \eta^{\text{dis}}$	0.95/0.95
AGV battery capacity	E_{cap}	100 kWh
Transportation energy per period	E_{consume}	5 kWh
AGV SoC range	$[SOC_{\min}, SOC_{\max}]$	[0.20, 1.00]
Initial AGV SoC	$SOC_{i,1}$	0.50
Available charging piles	N_{piles}	10
Required working AGVs	N_{demand}^t	3–14
Reefer-container number	N_{RC}	100
Maximum RC cooling power	$p_{\text{RC},\max}^{\text{RC}}$	50 kW
Initial RC temperature	$T_{r,1}$	−6 °C
Allowable RC temperature range	$[T_{r,\min}^{\text{min}}, T_{r,\max}^{\text{max}}]$	[−10, 0], °C
Thermal-inertia coefficient	α_r	0.80
Compressor coefficient of performance	η_r	3.0
Equivalent thermal resistance	R_r	1.0

TABLE VI
ENERGY-CONVERSION, P2G, AND CCS PARAMETERS IN THE CODE

Device	Capacity/limit	Efficiency/coefficient	Ramp limit
Grid	$P_{\text{grid},\max} = 4500$ kW	–	–
GT	$\bar{P}_{\text{GT},\max} = 4500$ kW	$(\eta_e, \eta_h) = (0.33, 0.42)$	250 kW
GT fuel input	$V_{\text{fuel},\max} = 52000$	$L^{\text{HV}} = 10$	–
HX	$P_{\text{HX},\max} = 1200$ kW	$\eta_{\text{HX}} = 0.80$	–
AC	$P_{\text{AC},\max} = 1000$ kW	$\eta_{\text{AC}} = 0.70$	–
HP	$P_{\text{HP},\max} = 1000$ kW	$\eta_{\text{HP}} = 0.96$	300 kW
ER	$P_{\text{ER},\max} = 1000$ kW	$\eta_{\text{ER}} = 1.50$	300 kW
EL	$P_{\text{EL},\max} = 300$ kW	$\eta^{\text{EL}} = 0.88$	60 kW
MR	$V_{\text{MR},\max} = 5000$	$(\eta^{\text{MR}}, \chi^{\text{MR}}) = (0.60, 0.75)$	–
HFC	$P_{\text{HFC},\max} = 250$ kW	$(\eta_e^{\text{HFC}}, \eta_h^{\text{HFC}}) = (0.95, 0.80)$	–
CCS base load	$P_{\text{CCS}}^b = 20$ kW	$\eta_{\text{CCS}} = 1.12$	–
CCS operating limit	$P_{\text{CCS},\max}^{\text{op},e} = 1000$ kW	$\chi_{\text{CCS}} = 0.40$	–
CCS total-power limit	$P_{\text{CCS},\max}^e = 1000$ kW	$\varphi_g = 2.0$	–
Captured CO ₂ limit	$E_{\max}^{\text{CCS}}$	–	–
Stored CO ₂ limit	$E_{\max}^{\text{S}}$	–	–

TABLE VII
TIME-SERIES INPUT RANGES USED BY THE CODE

Input profile	Manuscript symbol	24-h range
PV forecast	$\bar{P}_{\text{PV},t}^e$	0–1263 kW
WT forecast	$\bar{P}_{\text{WT},t}^e$	58.7–236.1 kW
VES electric-load forecast	$\bar{P}_{\text{VES},t}^e$	50–144 kW
Ambient temperature	T_t^{out}	6.0–18.0 °C
Port electric load	$P_{\text{tot},t}^e$	1320–3720 kW
Port heat load	$P_{\text{tot},t}^h$	420–1440 kW
Port cooling load	$P_{\text{tot},t}^c$	240–1680 kW